A Historical Development of Fluid Science

History of Fluid Mechanics, Meteorology & Oceanography

57 Important Scientists and Their Contributions
"From Ancient Hydrostatics to Modern Fluid Dynamics"

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and the help of OpenAI's ChatGPT Agent

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Aristotle (384–322 BC)

Aristotle was one of the most influential philosophers of ancient Greece, whose work covered natural philosophy, physics, biology, astronomy, logic, and meteorology.

Contribution to fluid mechanics, meteorology and oceanography:

Aristotle attempted to explain natural motion and the behaviour of fluids using observations of the physical world. In his work *Meteorologica*, he discussed winds, clouds, rain, evaporation, water circulation, storms, and other atmospheric phenomena.

He proposed that water could be transformed into vapour by heating and later return as precipitation. Although his explanations were not based on modern thermodynamics, this was an early attempt to describe the atmosphere as part of a continuous physical cycle.

Aristotle also recognized that air and water behave differently from solid materials and discussed the resistance offered by a surrounding medium to moving objects. His ideas were qualitative rather than mathematical, but they helped establish the study of fluids and the atmosphere as subjects of natural philosophy.

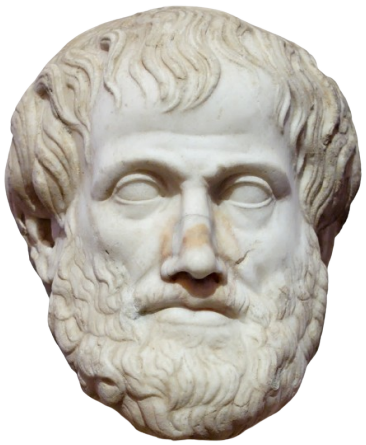


Figure 1: Aristotle of Stagira

Ctesibius of Alexandria (c. 285–222 BC)

Ctesibius was an ancient Greek engineer and inventor known for developing piston pumps, pneumatic devices, the hydraulic organ, and the water clock.

Contribution to fluid mechanics:

Ctesibius was one of the first people to systematically exploit the mechanical behaviour of air and water. He developed improved piston pumps in which a piston moved inside a cylinder and forced water through valves. This demonstrated that a mechanical device could use pressure differences to transport water.

He also developed the hydraulic organ, which used water to regulate the pressure of air supplied to pipes. His work represents an important early connection between mechanical motion, pressure, and fluid flow.

Ctesibius also worked with compressed air. His experiments helped establish that air could be compressed and used to transmit mechanical forces, an idea that later became fundamental to pneumatics and fluid power.

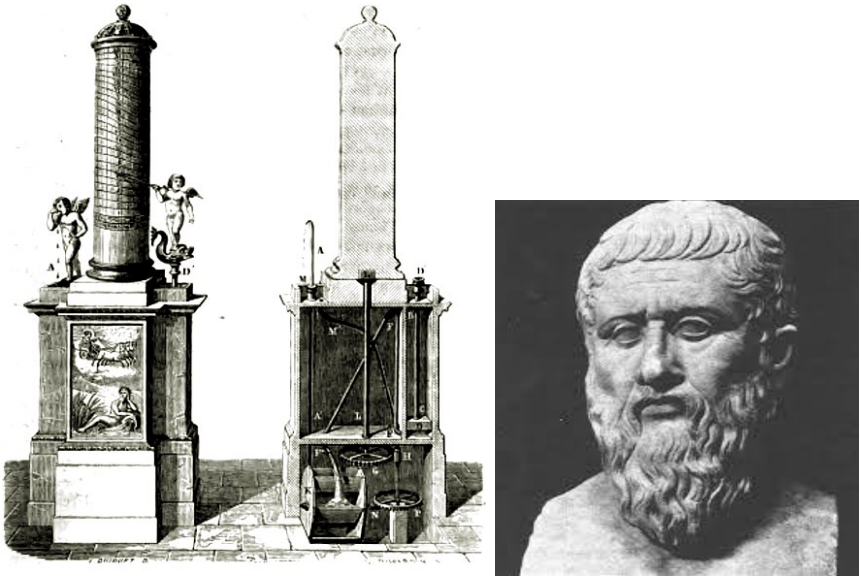


Figure 2: Ctesibius of Alexandrias water clock, illustrated by Claude Perrault.

Archimedes of Syracuse (c. 287–212 BC)

Archimedes was a Greek mathematician, physicist, engineer, and astronomer who established the foundations of hydrostatics and is famous for the Archimedes' principle of buoyancy.

Contribution to fluid mechanics:

Archimedes was a Greek mathematician, physicist, and engineer. He was educated in Alexandria and later returned to Syracuse, where he studied mechanics, geometry, hydrostatics, and practical machines such as the Archimedes' screw.

Archimedes established the foundations of hydrostatics through his study of floating and submerged bodies. In his work *On Floating Bodies*, he showed that a body immersed in a fluid experiences an upward force equal to the weight of the fluid displaced:

$$F_B = W_{\text{displaced}} = \rho_f g V_{\text{displaced}},$$

where F_B is the buoyant force, ρ_f is the fluid density, and $V_{\text{displaced}}$ is the displaced fluid volume. For a floating body, this upward force balances its weight,

$$F_B = W.$$

This explains why an object sinks only until enough fluid has been displaced to provide the required upward force. Archimedes also investigated the equilibrium and stability of floating bodies, making his work one of the earliest mathematical treatments of fluids at rest.

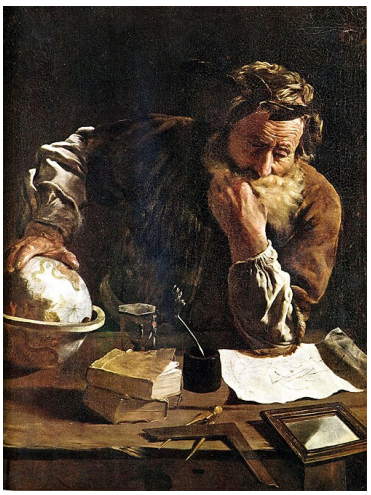


Figure 3: Archimedes of Syracuse by Domenico Fetti.

Hero of Alexandria (c. 10–70 AD)

Hero of Alexandria was a Greek-Egyptian mathematician and engineer known for pneumatic machines, automatic devices, the aeolipile, and early hydraulic engineering.

Contribution to fluid mechanics:

Hero investigated the behaviour of air and water in enclosed systems and designed machines that exploited pressure differences. His *Pneumatica* described devices using air pressure, water pressure, siphons, fountains, valves, and compressed air.

One of his most famous inventions was the aeolipile. Water was heated inside a vessel until steam escaped through bent nozzles. The escaping jets produced a reaction force that caused the vessel to rotate. Although the aeolipile was not developed into a practical steam engine, it demonstrated the mechanical effect of fluid jets. Hero's work is particularly important because it connected fluid pressure, fluid discharge, and mechanical motion in practical machines.

Figure 4: *Codex of Saint Gregory Nazianzus* including a drawing of Hero of Alexandria

Ibn al-Haytham (c. 965–c. 1040)

Ibn al-Haytham was an Iraqi mathematician and physicist best known for his pioneering work in optics, experimental science, and the study of light and vision.

Contribution to fluid mechanics:

Ibn al-Haytham studied the atmosphere as a physical medium and investigated how it affects the path of light. By studying atmospheric refraction and using geometrical reasoning, he made an early quantitative estimate of the height or thickness of the atmosphere.

Although his estimate was based on simplified assumptions and differs greatly from the modern structure of the atmosphere, it was significant because he attempted to determine a physical property of the atmosphere from observable effects. His work therefore represents an early connection between geometry, observation, and the physical study of the atmosphere, preceding the later quantitative development of atmospheric and fluid science.

Figure 5: Ibn al-Haytham by Jeremias Falck

Leonardo da Vinci (1452–1519)

Leonardo da Vinci was an Italian artist, engineer, and scientist who made some of the earliest detailed scientific observations of water waves, vortices, turbulence, and hydraulic flows.

Contribution to fluid mechanics:

Leonardo made remarkably detailed observations of moving water. He studied rivers, channels, waves, eddies, vortices, jets, hydraulic jumps, and the motion of water around obstacles.

He drew observations of water flowing through channels and recognized that turbulence contains complicated rotating structures. He also observed that water issuing from an opening forms a jet whose shape changes as it falls.

Leonardo investigated the relationship between the width and depth of flowing channels and the amount of water transported. He also studied waves and recognized that wave motion can propagate through water without the water particles themselves travelling with the wave over long distances.

His notebooks contain some of the earliest detailed scientific drawings of turbulent flow and hydraulic phenomena.

Figure 6: Water falling upon water by Leonardo da Vinci

Simon Stevin (1548–1620)

Simon Stevin was a Flemish mathematician and engineer who established important foundations of hydrostatics and statics.

Contribution to fluid mechanics:

Stevin developed an early mathematical theory of hydrostatics. He demonstrated that pressure in a fluid increases with depth and recognized that the pressure at a given depth depends on the depth and density of the fluid rather than the shape of the container.

This is expressed today as

$$p = p_0 + \rho g h.$$

This idea is sometimes illustrated with Stevin's "hydrostatic paradox": containers with very different shapes can produce the same pressure at the same depth.

Stevin's work helped establish the concept that fluid pressure is a local property of the fluid and does not simply depend on the total amount of fluid contained in a vessel.

Figure 7: Wind chariot or land yacht designed by Simon Stevin

Galileo Galilei (1564–1642)

Galileo Galilei was an Italian physicist and astronomer who pioneered experimental mechanics and quantitative studies of motion and floating bodies.

Contribution to fluid mechanics:

Galileo studied the behaviour of bodies immersed in fluids, especially the problem of why some objects float while others sink. He examined density and specific gravity and investigated floating bodies experimentally. His work challenged older explanations of buoyancy and helped move the subject toward quantitative measurement.

A particularly important problem was the apparent paradox of a heavy ship floating while a small piece of the same material sinks. Galileo's analysis helped establish that floating depends on the relationship between the body's weight and the hydrostatic forces acting on it, rather than simply on whether the material itself is heavier than water. His work also contributed to the experimental approach that later became central to fluid mechanics.

Figure 8: Galileo Galilei

Otto von Guericke (1602–1686)

Otto von Guericke was a German physicist and engineer who demonstrated the enormous force of atmospheric pressure through his vacuum experiments and the famous Magdeburg hemispheres.

Contribution to fluid mechanics and atmospheric science:

Von Guericke constructed one of the first effective air pumps and used it to investigate what happens when air is removed from a vessel. His most famous demonstration involved the Magdeburg hemispheres. Two metal hemispheres were joined and the air inside was pumped out. Teams of horses were unable to pull the hemispheres apart. Once air was allowed back inside, they could be separated easily.

The experiment demonstrated the enormous mechanical effect of atmospheric pressure. If the pressure difference across an area is Δp, the resulting force is approximately

$$F = \Delta p A.$$

Von Guericke's experiments provided powerful experimental evidence that air exerts pressure and that atmospheric pressure is a real mechanical force.

Figure 9: Gaspar Schott's sketch of Otto von Guericke's Magdeburg hemispheres experiment

Evangelista Torricelli (1608–1647)

Evangelista Torricelli was an Italian physicist and mathematician who invented the mercury barometer and derived the relation for the speed of liquid discharge from an opening.

Contribution to fluid mechanics:

Torricelli studied the speed at which a liquid leaves a tank through a small opening.

For a tank with a free surface at height h above the opening, he derived the result

$$v = \sqrt{2gh} \implies v \propto \sqrt{h}.$$

The result shows that the exit velocity is related to the conversion of gravitational potential energy into kinetic energy. It is an early form of what is now recognized as a Bernoulli-type energy relation.

Torricelli also invented the mercury barometer. A long glass tube was filled with mercury, inverted into a reservoir, and left with a vacuum above the mercury column. Atmospheric pressure supporting the mercury produced the barometric height.

The relation is

$$p_{\text{atm}} = \rho_{\text{Hg}} g h.$$

This connected atmospheric pressure to a measurable height and became fundamental to meteorology.

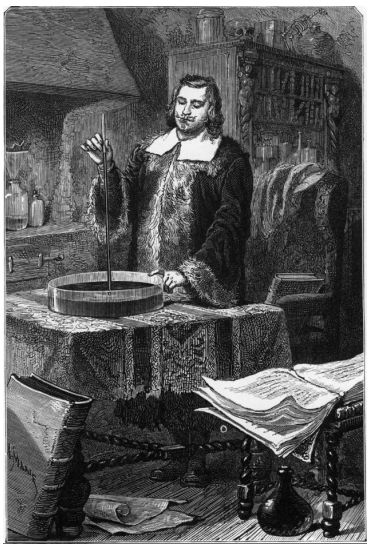


Figure 10: Evangelista Torricelli

Blaise Pascal (1623–1662)

Blaise Pascal was a French mathematician and physicist who established important principles of fluid pressure and demonstrated that atmospheric pressure decreases with altitude.

Contribution to fluid mechanics:

Pascal investigated atmospheric pressure and how pressure is transmitted through fluids. His experiments helped establish that the atmosphere exerts pressure because of the weight of the air above us. In 1648, his brother-in-law Florin Périer carried out Pascal's famous experiment on the Puy de Dôme, measuring the height of a mercury column at different elevations. The mercury column became shorter at higher altitude, showing that atmospheric pressure decreases with height.

Pascal also used a simple but striking argument involving a long tube filled with wine. If a tube is inverted into a container of wine, the atmosphere can support the column of liquid, but only up to a certain height. For a liquid column in hydrostatic equilibrium, gives a maximum height of roughly 10 m under normal atmospheric pressure. This was an important demonstration that it is the atmosphere pushing the liquid upward, rather than the act of "suction" pulling the liquid upward.

Pascal also established the principle now known as Pascal's law: a pressure change applied to a confined fluid is transmitted throughout the fluid. Since pressure is force over area one obtains:

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}.$$

These experiments and principles made Pascal one of the key figures in the development of hydrostatics and the understanding of atmospheric pressure.

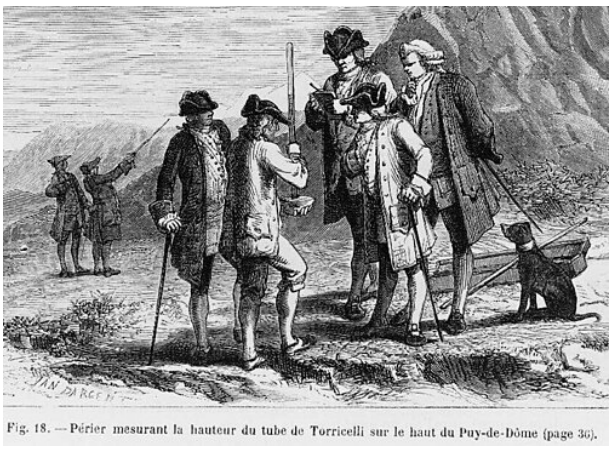


Fig. 18. — Périer mesurant la hauteur du tube de Torricelli sur le haut du Puy-de-Dôme (page 36).

Figure 11: Florin Périer on the Puy de Dôme

Robert Boyle (1627–1691)

Robert Boyle was an Irish physicist and chemist who pioneered the experimental study of gases and established the pressure–volume relationship known as Boyle's law.

Contribution to fluid mechanics and gas physics:

Boyle became interested in the air pump developed by Otto von Guericke and worked on improving the design of the pump. He and his collaborators used air pumps to investigate the properties of air and the behaviour of gases under reduced pressure. These experiments demonstrated that air is compressible and that its pressure and volume are quantitatively related. Boyle's work, together with earlier experiments by Richard Towneley, established the inverse relationship between the pressure and volume of a fixed amount of gas at approximately constant temperature, known as Boyle's law

$$pV = \text{constant} \implies p \propto \frac{1}{V},$$

Boyle also proposed a corpuscular theory of matter, according to which substances are composed of extremely small particles separated by empty space. He used this idea to explain the compressibility of air: when air is compressed, the particles themselves need not become smaller; rather, the spaces between them are reduced. This represented an important move away from the Aristotelian idea that matter was continuously divisible and toward a particulate description of gases. His experiments on vacuum, air pressure, combustion, and the physical properties of gases helped establish the experimental study of fluids and gases as a quantitative science.

Figure 12: *An Experiment on a Bird in the Air Pump* by Joseph Wright of Derby, displaying Robert Boyle

Robert Hooke (1635–1703)

Robert Hooke was an English physicist and polymath known for Hooke's law, microscopy, and early ideas about the microscopic motion of matter and gases.

Contribution to fluid mechanics and gas physics:

Hooke developed an early microscopic explanation of the behaviour of gases and thermal expansion. He proposed that air consists of very small particles in continual motion and that the impacts of these particles against surfaces produce air pressure. This is an important qualitative precursor to the kinetic theory of gases.

He also proposed that heating matter changes the motion of its microscopic particles. Increased molecular motion could cause particles to separate farther apart, explaining the expansion of matter when heated. This idea is particularly important because it connected temperature and heat with microscopic motion rather than treating heat as a separate material substance.

Hooke's ideas therefore anticipated two major concepts of later physics:

$$\text{gas pressure} \longleftrightarrow \text{microscopic particle motion},$$

$$\Longrightarrow \text{ thermal expansion} \longleftrightarrow \text{increased microscopic motion}.$$

These ideas later became central to kinetic theory and thermodynamics.

Figure 13: Portrait of a Mathematician by Mary Beale, could be Hooke

Isaac Newton (1643–1727)

Isaac Newton was an English mathematician and physicist who formulated the laws of motion and gravitation and established the constitutive relation for Newtonian viscosity.

Contribution to fluid mechanics:

Newton established the mathematical framework for applying forces to moving fluids. His second law,

$$\mathbf{F} = m\mathbf{a},$$

provides the basic momentum equation from which fluid dynamics can be developed.

Newton also investigated the resistance and internal friction of fluids. For a Newtonian fluid, the shear stress is proportional to the velocity gradient:

$$\tau = \mu \frac{du}{dy} \implies \tau \propto \frac{du}{dy}.$$

Here μ is the dynamic viscosity. This relationship is the defining constitutive relation for a Newtonian fluid and remains fundamental to modern fluid mechanics.

Newton also studied the motion of fluids in his *Principia*, including the resistance experienced by bodies moving through fluids.

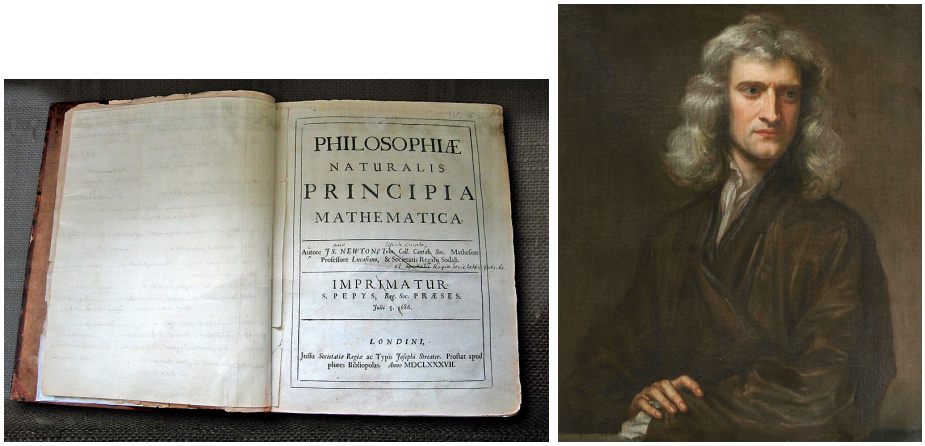


Figure 14: Isaac Newtons *Principia*

Edmond Halley (1656–1742)

Edmond Halley was an English astronomer and mathematician who produced one of the first global maps of prevailing winds and proposed an early theory of atmospheric circulation.

Contribution to meteorology and physical oceanography:

Halley was a remarkable scientific multitasker whose work ranged from astronomy and mathematics to geophysics, meteorology, navigation, and statistics. He mapped the southern stars, correctly predicted the return of the comet now known as Halley's Comet, studied Earth's magnetic field and produced an early magnetic-declination map, investigated tides and evaporation, and developed one of the earliest statistical life tables. He also designed and tested a diving bell, descending in the River Thames to demonstrate its use.

Halley then made one of the earliest serious attempts to describe global atmospheric circulation from observations. During his voyage to St. Helena, he studied the prevailing winds and combined his observations with reports from other parts of the world to produce the first published global chart of prevailing winds in 1686. He proposed that solar heating drives the large-scale circulation, with stronger heating near the equator causing air to rise and surrounding air to move toward the heated region. Although his circulation mechanism was incomplete, this was an important early attempt to construct a physical model of the atmosphere from global observations.

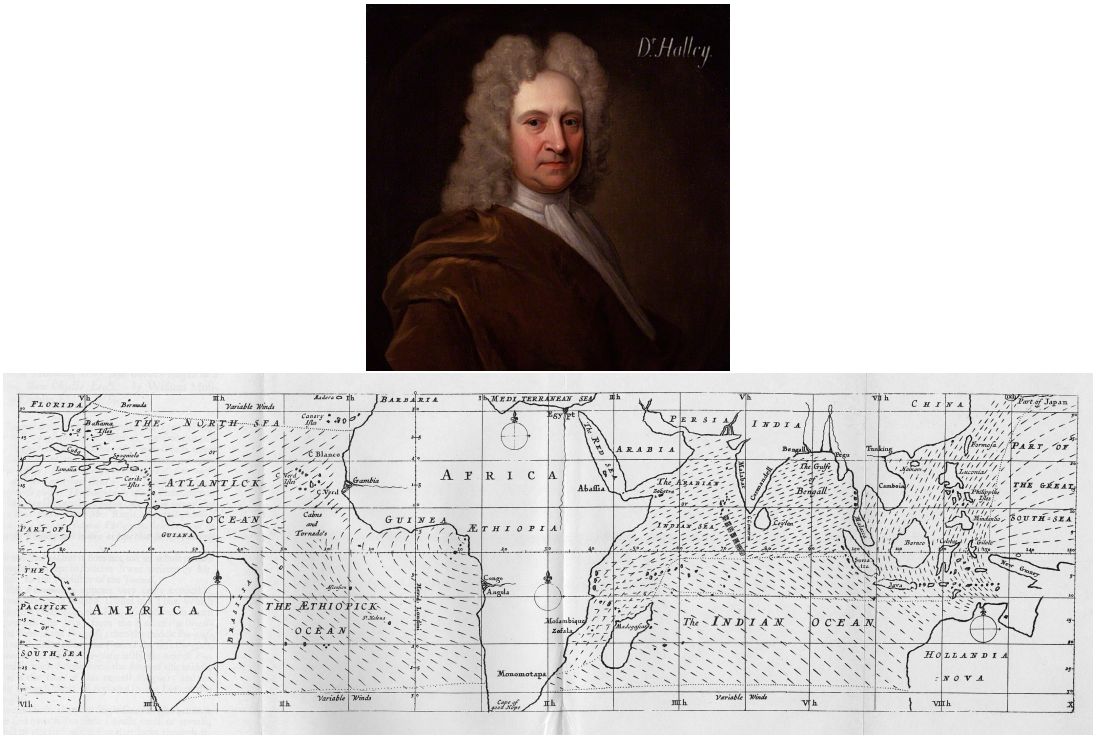


Figure 15: Edmond Halleys map showing large scale circulation

George Hadley (1685–1768)

George Hadley was an English meteorologist who gave the first convincing physical explanation of the trade winds and the large-scale tropical circulation now known as the Hadley circulation.

Contribution to meteorology:

Hadley was an amateur meteorologist who systematically studied meteorological observations collected by the Royal Society. He was elected a Fellow of the Royal Society in 1735 and that same year published his explanation of the trade winds. His work was initially little known and was later independently developed by John Dalton, who eventually recognised Hadley's priority. The theory gradually became known as Hadley's principle and later as the Hadley circulation. Hadley's brother, John Hadley, was also a prominent inventor who developed the octant, an important predecessor of the sextant for navigation.

In his 1735 paper, Hadley argued that strong solar heating near the equator causes warm air to rise and flow poleward at high altitude, while cooler air returns toward the equator near the surface. He recognised that Earth's rotation deflects these air currents, helping explain the direction of the trade winds. Although his theory was incomplete, this was an important early attempt to explain global atmospheric circulation as the combined result of unequal heating and planetary rotation.

The circulation pattern associated with Hadley's theory is now called the Hadley circulation and forms the tropical branch of the Earth's large-scale meridional atmospheric circulation.

Figure 16: George Hadley

Daniel Bernoulli (1700–1782)

Daniel Bernoulli was a Swiss mathematician and physicist who developed the energy principle for flowing fluids now expressed by Bernoulli's equation.

Contribution to fluid mechanics:

Bernoulli's most important work on fluids was published in *Hydrodynamica* in 1738. He developed the first mathematical description of fluid motion based on the conservation of *vis viva*, an early form of the modern conservation of kinetic and potential energy. For steady, incompressible, inviscid flow along a streamline, this gives

$$p + \frac{1}{2}\rho v^2 + \rho g z = \text{constant}.$$

The three terms represent pressure energy per unit volume, kinetic energy per unit volume, and gravitational potential energy per unit volume. At approximately constant elevation,

$$p + \frac{1}{2}\rho v^2 = \text{constant},$$

so an increase in flow speed is accompanied by a decrease in static pressure.

Bernoulli also worked on vibrating strings, mechanics, and the theory of tides, contributing to one of the major eighteenth-century attempts to develop mathematical descriptions of physical systems. His *Hydrodynamica* helped establish fluid mechanics as a quantitative science based on conservation principles.

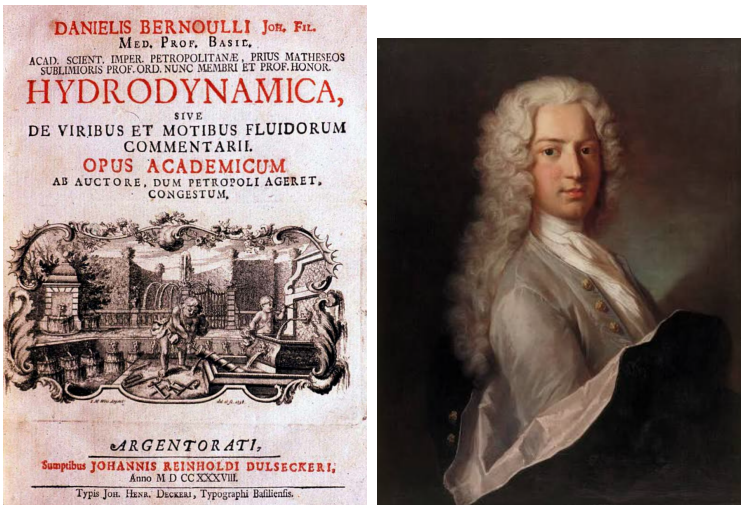


Figure 17: *Hydrodynamica* by Daniel Bernoulli

Leonhard Euler (1707–1783)

Leonhard Euler was a Swiss mathematician who formulated the differential equations governing ideal fluid motion and helped establish the mathematical foundations of fluid dynamics.

Contribution to fluid mechanics:

Euler formulated the differential equations governing ideal fluid motion. For an inviscid fluid, the momentum equation can be written as

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \rho\mathbf{g}.$$

These are now called the Euler equations. Euler's equations describe the acceleration of a fluid resulting from pressure gradients and body forces when viscosity is neglected.

Euler also introduced the modern mathematical description of fluid velocity as a field,

$$\mathbf{u} = \mathbf{u}(\mathbf{x}, t),$$

allowing fluid motion to be described mathematically at every point in space and time.

Figure 18: Leonhard Euler

Jean le Rond d'Alembert (1717–1783)

Jean le Rond d'Alembert was a French mathematician and physicist who identified the ideal-fluid paradox of zero drag on a body in steady inviscid flow.

Contribution to fluid mechanics:

D'Alembert investigated the motion of ideal fluids around solid bodies. His analysis led to what became known as d'Alembert's paradox: for steady, inviscid, incompressible, irrotational flow around a body, the ideal-fluid equations predict zero net drag.

This was puzzling because real objects moving through fluids clearly experience drag. The paradox became historically important because it demonstrated that an apparently reasonable ideal-fluid model was missing an essential physical mechanism.

The missing physics is primarily viscosity and the resulting boundary layer, flow separation, and wake.

Figure 19: Jean le Rond d'Alembert

Joseph-Louis Lagrange (1736–1813)

Joseph-Louis Lagrange was an Italian-French mathematician and physicist who developed analytical mechanics and mathematical methods for describing motion using energy principles and generalized coordinates.

Contribution to fluid mechanics:

Lagrange developed mathematical methods for describing mechanics using generalized coordinates and energy principles. His work helped establish the distinction between following an individual particle and describing the entire velocity field.

In a Lagrangian description of a fluid, one follows an individual fluid particle:

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}, t).$$

This approach remains fundamental to particle-based descriptions of fluid motion and is especially useful when following water parcels in oceanography. Lagrange's analytical mechanics also provided mathematical tools later used in fluid dynamics and geophysical fluid mechanics.

Figure 20: Joseph-Louis Lagrange

Pierre-Simon Laplace (1749–1827)

Pierre-Simon Laplace was a French mathematician and physicist who was ahead of his time and made major contributions to mathematical physics, astronomy, probability, and the theory of sound. He also formulated the philosophical deterministic *Laplaces Demon*.

Contribution to fluid mechanics and geophysics:

Laplace was a professor at the École Militaire and was elected to the French Academy of Sciences in 1773. He became a leading scientific figure in France and briefly served as Minister of the Interior under Napoleon. His five-volume *Mécanique céleste* (1799–1825) synthesized the work of Newton, Euler, d'Alembert, and Lagrange.

Laplace investigated the propagation of sound in gases and corrected Newton's treatment. He recognized that rapid compression and expansion are approximately adiabatic. The speed of sound in an ideal gas is

$$c = \sqrt{\frac{\gamma p}{\rho}},$$

where γ is the ratio of specific heats. Laplace also developed the mathematical theory of tides on the rotating Earth. Using spherical coordinates, he formulated the *Laplace tidal equations*, accounting for Earth's spherical geometry and the Coriolis effect. These equations became fundamental to atmospheric and oceanic wave theory. Later developments revealed two major wave classes: high-frequency inertia–gravity waves, associated with Poincaré waves, and low-frequency planetary waves, including Rossby waves. Laplace also developed potential theory and differential equations. The Laplace equation,

$$\nabla^2 \phi = 0,$$

and the Laplacian operator became fundamental in physics. His work in probability and statistics, including least-squares methods, provided important tools for analysing scientific observations. Laplace therefore helped establish mathematical physics as a framework for describing sound, tides, atmospheric motion, ocean waves, and planetary systems.

Figure 21: Pierre-Simon Laplace

Claude-Louis Navier (1785–1836)

Claude-Louis Navier was a French engineer and mathematician who introduced viscous stresses into the mathematical equations of fluid motion, helping establish the Navier–Stokes equations.

Contribution to fluid mechanics:

Navier extended the equations of fluid motion to include internal friction. The resulting equation has the form

$$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \rho\mathbf{g},$$

for an incompressible Newtonian fluid with constant viscosity. The viscous term

$$\mu\nabla^2\mathbf{u}$$

represents the diffusion of momentum caused by molecular interactions. Navier's work was one of the steps leading toward the modern Navier–Stokes equations.

Figure 22: Claude-Louis Navier

Gaspard-Gustave de Coriolis (1792–1843)

Gaspard-Gustave de Coriolis was a French mathematician and engineer who formulated the effects of rotation on moving bodies that are now known as the Coriolis force.

Contribution to fluid mechanics and geophysics:

Gaspard-Gustave de Coriolis was a French mathematician and engineer who worked at the École Polytechnique in Paris, where he studied mechanics and rotating systems. Interestingly, it was not the atmosphere nor ocean which made Coriolis interested in the subject, but billiard. The rector of the École Polytechnique was an enthusiastic billiards player and wanted to understand how billiard balls rolled and bounced. He therefore asked Coriolis to investigate the mathematics of the game, leading to Coriolis' 1835 work on the mathematical theory of billiard *Théorie mathématique du jeu de billard*.

Coriolis wanted to understand why some billiard balls follow a curved path when the rotate. He gave a mathematical description of the apparent forces that arise when motion is viewed from a rotating reference frame. He never considered this force for the atmosphere, but it was formalized as the Coriolis acceleration

$$\mathbf{a}_C = -2\boldsymbol{\Omega} \times \mathbf{u}.$$

where Ω angular rotation rate. The sinusoidal latitude dependence of Ω for the Globe was independently discoverd by Foucault and his pendulum. This term was later revealed to be essential for understanding large-scale atmospheric and oceanic circulation. It explains why winds and ocean currents are deflected relative to the direction of the pressure-gradient force: to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.

Figure 23: Gaspard-Gustave de Coriolis

George Gabriel Stokes (1819–1903)

George Gabriel Stokes was an Irish-British mathematician and physicist known for Stokes' law, viscous flow theory, and major contributions to mathematical physics.

Contribution to fluid mechanics:

Stokes was a British mathematician and physicist who worked at the University of Cambridge, where he became Lucasian Professor of Mathematics. He made major contributions to mathematical physics, fluid mechanics, optics, and wave theory. In fluid mechanics, Stokes studied the motion of very small particles through viscous fluids and developed what is now known as Stokes' law.

For a small spherical particle moving slowly through a viscous fluid, the drag force is

$$F_D = 6\pi\mu r v \implies F_D \propto v.$$

This is valid in the creeping-flow regime, where inertial effects are negligible. Stokes' law is used to determine the settling velocities of particles, droplets, aerosols, and sediment grains.

Balancing gravitational settling against viscous drag gives the terminal settling speed of a small sphere:

$$v_t = \frac{2r^2(\rho_p - \rho_f)g}{9\mu}.$$

This result is important in sediment transport, atmospheric aerosols, ocean particles, and laboratory fluid mechanics. Stokes also made fundamental contributions to the mathematical description of viscous flow, including the equations governing slow viscous motion that form an important part of modern fluid mechanics.

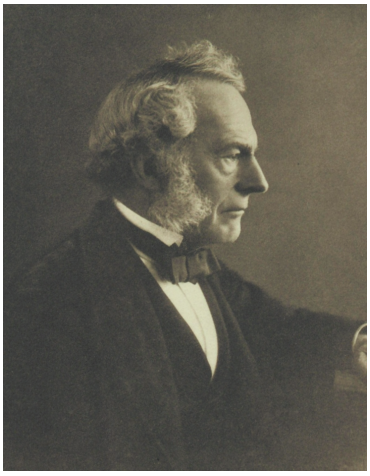


Figure 24: George Gabriel Stokes

Benoît Paul Émile Clapeyron (1799–1864)

Benoît Paul Émile Clapeyron was a French physicist and engineer who developed the mathematical theory of phase changes and the Clapeyron equation.

Contribution to atmospheric and thermodynamic physics:

Clapeyron developed a mathematical relationship describing phase changes. The general Clapeyron equation is

$$\frac{dp}{dT} = \frac{L}{T\Delta v},$$

where L is latent heat and Δv is the change in specific volume during the phase transition. For water vapour in the atmosphere, the relationship becomes approximately

$$\frac{d\ln p_{\text{sat}}}{dT} \approx \frac{L}{R_v T^2}.$$

This relationship explains why the saturation vapour pressure of water increases rapidly with temperature. That fact is fundamental to cloud formation, atmospheric humidity, evaporation, condensation, and weather systems.

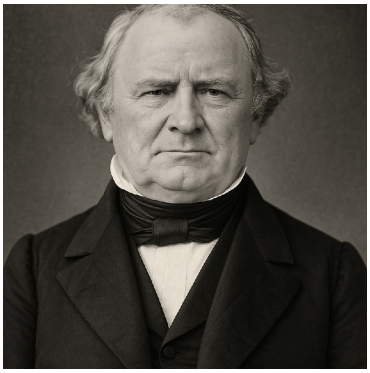


Figure 25: Benoît Paul Émile Clapeyron

Rudolf Clausius (1822–1888)

Rudolf Julius Emanuel Clausius was a German physicist who established the concept of entropy and helped formulate the second law of thermodynamics.

Contribution to atmospheric and fluid physics:

Clausius introduced the modern concept of entropy and formulated one of the central mathematical statements of the second law of thermodynamics. For a reversible process,

$$dS = \frac{\delta Q_{\text{rev}}}{T}.$$

Entropy became essential for describing irreversible processes and the thermodynamic behaviour of gases and fluids.

Clausius also developed the Clausius–Clapeyron equation, which describes how saturation vapour pressure changes with temperature:

$$\frac{d \ln p_{\text{sat}}}{dT} \approx \frac{L}{R_v T^2}.$$

This relation is central to meteorology because atmospheric water vapour, condensation, evaporation, clouds, and precipitation are controlled by thermodynamic phase changes.

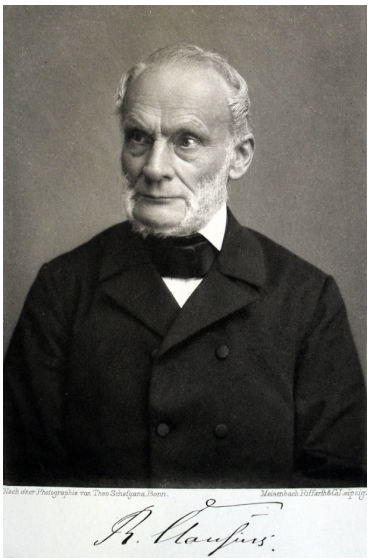


Figure 26: Rudolf Julius Emanuel Clausius

William Thomson, Lord Kelvin (1824–1907)

William Thomson, Lord Kelvin, was a Scottish-Irish physicist who made fundamental contributions to thermodynamics, wave theory, rotating flows, and the temperature scale.

Contribution to fluid mechanics and oceanography:

William Thomson, later known as Lord Kelvin, was a professor at the University of Glasgow, where he spent most of his scientific career and carried out major work in mathematical physics, thermodynamics, fluid mechanics, and geophysics. He developed the circulation theorem now known as Kelvin's circulation theorem. For an inviscid, barotropic fluid subject to conservative body forces,

$$\frac{D\Gamma}{Dt} = 0, \text{ where } \Gamma = \oint_C \mathbf{u} \cdot d\mathbf{l}$$

is the circulation around a material closed curve. This result became a foundation for the study of vorticity and rotating flows.

Kelvin also studied waves and developed theory relevant to water waves and tidal motion. His work on rotating fluids contributed to the mathematical understanding of large-scale geophysical flows, while his research into the Earth's physical properties connected mechanics with geophysics.

Beyond fluid mechanics, Kelvin was one of the major figures in nineteenth-century thermodynamics. He established the absolute temperature scale, formulated the second law of thermodynamics, and investigated the relationships between heat, work, and energy. He also made important contributions to electricity and magnetism, including electrical measurement, submarine telegraphy, and the description of physical fields. His broad work helped establish mathematical foundations that later became important for understanding the atmosphere, ocean, and Earth as interacting physical systems.

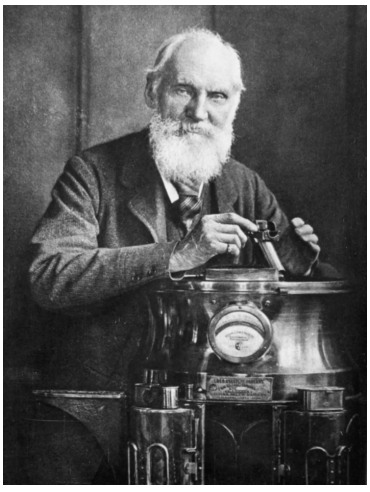


Figure 27: William Thomson, Lord Kelvin

Henri Poincaré (1854–1912)

Henri Poincaré was a French mathematician and physicist whose work on dynamical systems, nonlinear motion, and rotating-fluid waves laid foundations for modern chaos theory, special relativity and geophysical fluid dynamics.

Contribution to fluid mechanics:

Henri Poincaré worked at the University of Paris and was considered the most important French mathematician at the time. He is often described as the "last universal genius", because of the extraordinary breadth of his work across mathematics, physics, astronomy, and philosophy. Poincaré also worked on the emerging theory of relativity around the same period as Albert Einstein. He independently developed important elements of relativistic physics and helped clarify the mathematical structure of Lorentz transformations, while Einstein developed the special theory of relativity in 1905.

Poincaré's study of dynamical systems and celestial mechanics showed that deterministic systems can develop extremely complicated behaviour and become highly sensitive to their initial conditions. His work on the three-body problem revealed structures that later became fundamental to chaos theory and nonlinear dynamics.

Poincaré also studied waves in rotating fluids. His work on the shallow-water equations led to what are now called Poincaré waves, or inertia-gravity waves, in which gravity and the Coriolis effect interact. These waves are important in both atmospheric and oceanic flows.

He also introduced the Poincaré recurrence theorem and developed geometric tools such as Poincaré sections for analysing complex dynamical systems. His ideas laid important foundations for modern chaos theory, nonlinear dynamics, and the mathematical study of atmospheric and oceanic predictability.

Figure 28: Henri Poincaré

James Clerk Maxwell (1831–1879)

James Clerk Maxwell was a Scottish physicist and mathematician who unified electromagnetism and made fundamental contributions to kinetic theory, gas viscosity, and statistical physics.

Contribution to fluid and gas physics:

Maxwell made major contributions to the kinetic theory of gases, building on earlier work by Daniel Bernoulli, James Joule, and Rudolf Clausius. He described gases as large collections of rapidly moving molecules whose collisions produce macroscopic properties such as pressure and temperature.

Between 1859 and 1866, Maxwell developed the statistical distribution of molecular velocities. This became known as the Maxwell distribution, later generalized by Ludwig Boltzmann as the Maxwell–Boltzmann distribution. It describes the fraction of gas molecules moving at different speeds for a given temperature.

Maxwell also investigated the viscosity of gases and showed how molecular motion transports momentum between neighbouring layers. This provided a microscopic explanation for viscosity and helped establish the broader idea that momentum, heat, and mass can be transported through a fluid by molecular motion.

His work on thermodynamics also led to the Maxwell relations and the famous Maxwell's demon thought experiment concerning the second law of thermodynamics. His kinetic theory provided an important bridge between microscopic molecular physics and macroscopic fluid mechanics.

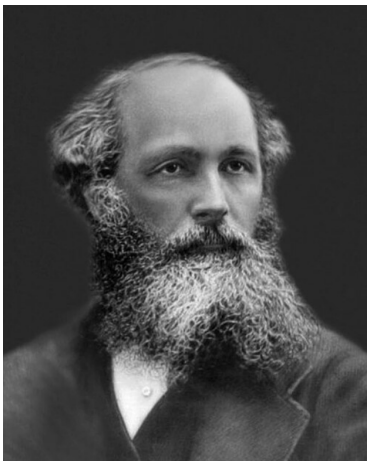


Figure 29: James Clerk Maxwell

Ludwig Boltzmann (1844–1906)

Ludwig Boltzmann was an Austrian physicist who established the statistical interpretation of thermodynamics and developed the kinetic theory and statistical mechanics of gases.

Contribution to fluid mechanics and climate physics:

Boltzmann developed the statistical description of gases, showing how macroscopic properties such as pressure, temperature, and entropy arise from molecular motion. His work extended Maxwell's kinetic theory and led to the Maxwell–Boltzmann distribution of molecular velocities.

His Boltzmann equation describes how the molecular distribution changes through motion and collisions,

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f + \vec{a} \cdot \nabla_{\vec{v}} f = \left(\frac{\partial f}{\partial t} \right)_{\text{coll}},$$

providing a fundamental link between molecular physics and macroscopic properties such as viscosity, diffusion, and heat conduction.

Boltzmann also gave entropy a statistical interpretation,

$$S = k_B \ln \Omega,$$

and provided the theoretical basis for the Stefan–Boltzmann law of thermal radiation, important for atmospheric and climate physics. After years of health problems and intense opposition to his atomic theory, Boltzmann died by suicide in 1906.

Figure 30: Ludwig Boltzmann

William Ferrel (1817–1891)

William Ferrel was an American meteorologist and self made scientist who developed the theory of midlatitude atmospheric circulation and explained the large-scale role of Earth's rotation.

Contribution to meteorology:

Ferrel was born into simple conditions in West Virginia in the United States. He came from a poor background but had a strong love for books and science. At 15, he worked in his neighbour's field all summer to earn enough money to buy a mathematics book. He later bought more books, continued studying, and eventually became a researcher in Missouri. He is best known as a meteorologist and mathematician who developed Hadley's ideas into a more physically accurate theory of atmospheric circulation. Unlike Hadley, Ferrel worked directly on meteorological problems and became one of the first scientists regarded as a theoretical meteorologist.

Ferrel recognized that Hadley's treatment of moving air was incomplete because it considered conservation of linear momentum rather than angular momentum about Earth's axis. An air mass moving relative to Earth tends to conserve angular momentum, changing its east–west velocity as it moves between latitudes. This helped explain the midlatitudes westerlies and the circulation of air in both hemispheres. His work contributed to what became known as Ferrel's law and the modern understanding of the Coriolis effect.

He also developed mathematical ideas about tides, despite apparently having little direct experience observing the ocean. His work also contributed to the concept of the Ferrel cell, the midlatitudes circulation cell between the tropical Hadley cell and the polar cell. His theory therefore provided an important extension of Hadley's model and helped establish the modern three-cell picture of global atmospheric circulation.

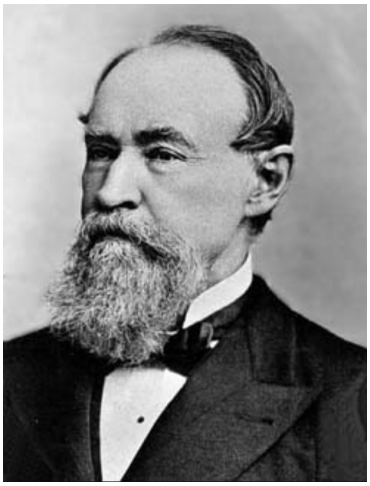


Figure 31: William Ferrel

Joseph Boussinesq (1842–1929)

Joseph Valentin Boussinesq was a French mathematician and physicist who developed the Boussinesq approximation and made major contributions to convection and turbulence theory.

Contribution to fluid mechanics and geophysical flows:

Boussinesq developed the approximation that bears his name. In many flows, density variations are small and can be ignored almost everywhere except where gravity converts those density differences into buoyancy. Thus, $\rho \approx \rho_0$ in most terms of the momentum equation, while density variations are retained in the buoyancy term. This approximation is fundamental to convection, atmospheric flows, ocean circulation, and laboratory experiments involving temperature-driven motion.

Boussinesq also developed an important approach to turbulent transport in which turbulent fluxes are related to mean gradients. A simple gradient-diffusion representation is

$$F_\phi = -K\frac{d\phi}{dz}.$$

For turbulent momentum transport this leads schematically to

$$\tau_t \approx \rho K_m \frac{dU}{dz},$$

with the precise sign depending on the coordinate and stress convention. These ideas became foundational for modelling turbulent geophysical flows.

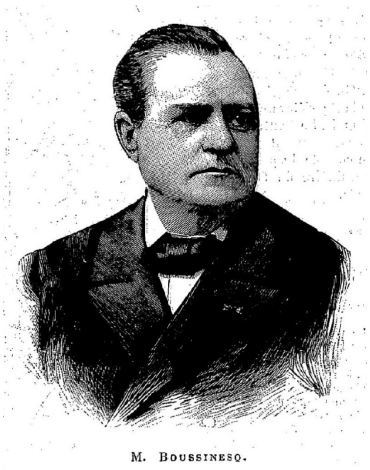


Figure 32: Joseph Valentin Boussinesq

John William Strutt, 3rd Baron Rayleigh (1842–1919)

Lord Rayleigh was an English physicist and mathematician who made fundamental contributions to wave theory, fluid instability, convection, and acoustics.

Contribution to fluid mechanics:

Rayleigh studied fluid instability and convection. One of the key dimensionless numbers associated with buoyancy-driven flow is the Rayleigh number:

$$Ra = \frac{g\alpha\Delta T L^3}{\nu\kappa},$$

where g is gravitational acceleration, α is thermal expansion coefficient, ΔT is the temperature difference, L is the characteristic length, ν is kinematic viscosity, and κ is thermal diffusivity. Large Ra means buoyancy is strong compared with viscous and thermal-diffusion effects, making convection more likely.

Rayleigh also investigated waves and instabilities, helping establish the mathematical treatment of instability that became central to modern fluid dynamics.

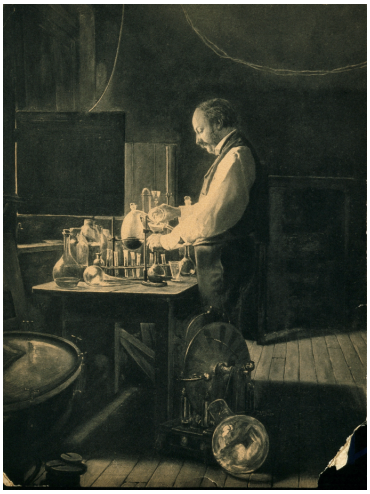


Figure 33: John William Strutt, 3rd Baron Rayleigh

Osborne Reynolds (1842–1912)

Osborne Reynolds was an English engineer and physicist who established the distinction between laminar and turbulent flow and introduced the Reynolds number.

Contribution to fluid mechanics:

Reynolds experimentally investigated the transition between smooth laminar flow and irregular turbulent flow. In his famous pipe-flow experiments, he injected dye into flowing water. At low flow speeds, the dye remained in a narrow, coherent streak. At higher speeds, the dye rapidly spread across the pipe as the flow became turbulent. He identified the dimensionless Reynolds number

$$Re = \frac{\rho U L}{\mu} = \frac{U L}{\nu}.$$

It compares inertial effects with viscous effects. Small Re generally corresponds to viscous-dominated flow, while large Re indicates that inertial effects are more important and turbulence becomes possible.

Reynolds also introduced the decomposition

$$u = U + u',$$

where U is the mean velocity and u' is the fluctuating component. This led to the Reynolds-averaged Navier–Stokes equations and the concept of turbulent stresses.

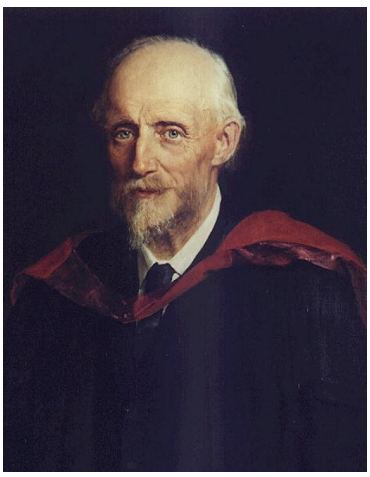


Figure 34: Osborne Reynolds

Matthew Fontaine Maury (1806–1873)

Matthew Fontaine Maury was an American naval officer and hydrographer who pioneered systematic ocean-current and wind charts for scientific navigation.

Contribution to physical oceanography:

Maury collected enormous quantities of observations from ships travelling across the oceans. He organized ship logs containing measurements of winds, currents, temperatures, and sailing conditions and converted them into systematic charts. These charts allowed navigators to identify recurring wind patterns and ocean currents and choose faster sailing routes.

Maury's work was important because it transformed scattered observations into a large-scale observational description of the oceans. His efforts also helped establish international cooperation in collecting oceanographic observations.

Figure 35: Matthew Fontaine Maury

Charles Wyville Thomson (1830–1882)

Charles Wyville Thomson was a Scottish marine scientist who led the Challenger expedition and helped establish systematic deep-sea oceanography.

Contribution to physical oceanography:

Thomson was the scientific leader of the *Challenger* expedition from 1872 to 1876. The expedition travelled around the world's oceans and collected systematic measurements of depth, temperature, seawater properties, sediments, and marine life. The expedition demonstrated that the deep ocean was not biologically dead and that ocean conditions varied systematically with depth and location.

Its measurements provided one of the first global observational datasets for physical and biological oceanography.

Figure 36: Charles Wyville Thomson

Sir John Murray (1841–1914)

Sir John Murray was a Scottish oceanographer who helped establish modern oceanography through the study of deep-sea sediments, ocean circulation, and marine environments.

Contribution to physical oceanography:

Murray participated in the *Challenger* expedition and later analyzed enormous quantities of oceanographic data. He investigated ocean depth, marine sediments, temperature distributions, ocean circulation, and the formation of deep-sea deposits.

He recognized that the oceans should be studied as interconnected physical systems rather than as isolated collections of biological organisms. Murray's work helped establish physical oceanography as a distinct scientific discipline.

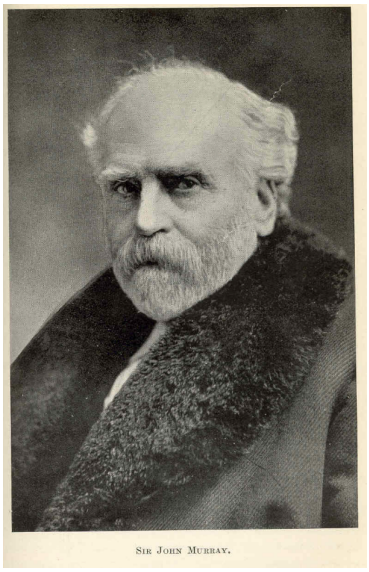


Figure 37: Sir John Murray

Fridtjof Nansen (1861–1930)

Fridtjof Nansen was a Norwegian polar explorer and scientist whose Fram expedition revealed important features of Arctic sea ice, stratification, and ocean circulation.

Contribution to physical oceanography:

Nansen's most important oceanographic work came from the *Fram* expedition of 1893–1896. The ship was specially designed with a rounded hull so that, when frozen into the Arctic sea ice, the pressure of the moving ice would lift the ship rather than crush it. Nansen deliberately allowed the *Fram* to freeze into the ice and drift with it across the Arctic Ocean.

During the expedition, Nansen measured temperature and salinity at different depths and identified distinct water masses beneath the cold surface layer. These observations helped establish that the Arctic Ocean contained warmer Atlantic water at depth, separated from the surface by strong stratification.

Nansen also noticed that the movement of the sea ice was systematically deflected relative to the wind rather than moving directly downwind. This observation provided an important experimental clue for Vagn Walfrid Ekman's later theory of wind-driven ocean transport and the Ekman spiral. Nansen's Arctic observations therefore helped connect sea-ice motion, ocean stratification, wind forcing, and large-scale ocean circulation, making the *Fram* expedition a landmark in modern physical oceanography.

Figure 38: Fridtjof Nansen

Vilhelm Bjerknes (1862–1951)

Vilhelm Bjerknes was a Norwegian physicist and meteorologist who established the mathematical foundation of modern weather prediction and founded the Bergen school of meteorology.

Contribution to meteorology:

Bjerknes recognized that weather prediction could be treated as an initial-value problem in fluid dynamics and thermodynamics. In principle, if the atmospheric variables could be measured sufficiently accurately at one time, their future evolution could be calculated from the governing physical laws. These variables include:

- $\vec{u}$ – velocity field
- p – atmospheric pressure
- T – temperature
- ρ – air density
- q – water-vapour content (or salinity for the ocean)

These variables are linked through the equations of fluid motion, conservation of mass, thermodynamics, the equation of state, and moisture processes. Bjerknes therefore helped transform meteorology from a largely descriptive science into a quantitative physical science.

Bjerknes also founded the Bergen school of meteorology, where these ideas were developed further through the study of atmospheric fronts, cyclones, air masses, and weather evolution. His work provided an important foundation for modern weather forecasting and atmospheric dynamics.

Figure 39: Vilhelm Bjerknes

Vagn Walfrid Ekman (1874–1954)

Vagn Walfrid Ekman was a Swedish oceanographer who explained wind-driven ocean transport through the Coriolis force, friction, and the Ekman spiral.

Contribution to physical oceanography:

Ekman became interested in oceanography after attending Vilhelm Bjerknes' lectures on fluid dynamics. He later worked with Bjerknes in Stockholm, where he came into contact with the observations from Fridtjof Nansen's *Fram* expedition. Nansen had observed that sea ice and icebergs drifted at an angle to the prevailing wind rather than directly with it. Bjerknes asked Ekman to investigate this phenomenon.

Ekman explained the deflection as a balance between wind stress, frictional momentum transfer, and the Coriolis force. His theory produced the Ekman spiral, in which both the speed and direction of current change with depth. The vertically integrated Ekman transport is

$$\mathbf{M}_E = \frac{1}{\rho f}\hat{\mathbf{z}} \times \boldsymbol{\tau},$$

where $\boldsymbol{\tau}$ is wind stress, ρ is seawater density, and f is the Coriolis parameter. The resulting transport is approximately 90° to the wind stress, with its direction depending on the hemisphere.

Ekman also developed instruments such as the Ekman current meter and water bottle, making important contributions to the experimental measurement of ocean currents and water properties. His work provided a fundamental explanation of how atmospheric winds drive ocean circulation.

Figure 40: Vagn Walfrid Ekman

David Brunt (1886–1965)

David Brunt was a British meteorologist and mathematician known for his work on atmospheric stability and the Brunt–Väisälä frequency.

Contribution to fluid mechanics:

Brunt studied the vertical motion of air parcels in a stratified atmosphere. He showed that when an air parcel is displaced vertically, differences between its density and the surrounding air can produce a restoring force. For stable stratification, the parcel oscillates about its equilibrium position according to

$$\frac{d^2\xi}{dt^2} + N^2\xi = 0,$$

where N is the buoyancy frequency. The quantity N is now known as the Brunt–Väisälä frequency,

$$N^2 = \frac{g}{\theta}\frac{d\theta}{dz}.$$

A positive N^2 indicates stable stratification, while the magnitude of N measures the strength of the restoring effect. This concept became fundamental in the study of atmospheric and oceanic stratification, internal waves, convection, and vertical mixing.

In 1926, Brunt published *Periodicities in European Weather*, in which he investigated recurring patterns in European weather and their possible physical causes, contributing to the early development of statistical and dynamical approaches to weather prediction.

Figure 41: David Brunt

Wasaburo Oishi (1874–1950)

Wasaburo Oishi was a Japanese meteorologist who was one of the first to find the jet stream.

Contribution to meteorology:

Oishi's balloon observations revealed strong, persistent westerly winds high above Japan. His measurements, published in the 1920s, provided some of the earliest systematic evidence of what became known as the jet stream. Oishi also helped establish the importance of upper-air measurements for understanding and forecasting the weather. Interestingly, he published much of his research in Esperanto, an international auxiliary language, hoping to make his findings accessible to scientists worldwide. This language choice, however, meant that his work received limited attention in the English-speaking scientific community for many years.

During the Second World War, Japan launched thousands of Fu-Go balloon bombs, carrying incendiary and explosive devices across the Pacific towards North America. These balloons exploited the jet stream. There is an intriguing historical irony: had more American scientists been able to read Oishi's Esperanto reports, they might have learned about the jet stream earlier and better understood the atmospheric mechanism behind the balloon attacks.

Figure 42: Wasaburo Oishi

Ludwig Prandtl (1875–1953)

Ludwig Prandtl was a German physicist and engineer who founded boundary-layer theory and pioneered modern approaches to turbulence modelling.

Contribution to fluid mechanics:

Prandtl developed boundary-layer theory, showing that viscosity is concentrated in a thin region near a solid surface while the outer flow can often be treated as nearly inviscid. This explained drag and flow separation and resolved the difficulty posed by d'Alembert's paradox.

He also developed the mixing-length hypothesis for turbulent momentum transport. He proposed that turbulent eddies transport fluid parcels over a characteristic distance l_m, producing an additional turbulent momentum flux. The turbulent shear stress can be approximated by

$$\tau_t \approx \rho l_m^2 \left|\frac{dU}{dz}\right| \frac{dU}{dz}.$$

This idea became an important foundation for later turbulence models and atmospheric boundary-layer theory.

Prandtl also made major contributions to wing theory, induced drag, wing-tip vortices, and supersonic flow. His lifting-line theory showed how the lift distribution along a finite wing affects induced drag. Theodore von Kármán was one of his students at Göttingen and later became a major figure in aerodynamics and turbulence.

During the Nazi period, Prandtl remained a leading figure in German science and expressed support for National Socialism in correspondence.

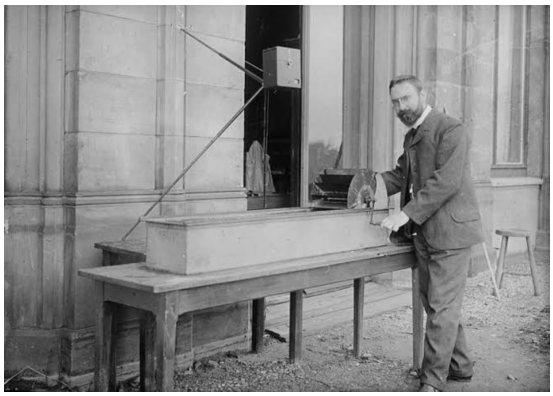


Figure 43: Ludwig Prandtl is shown next to a water channel

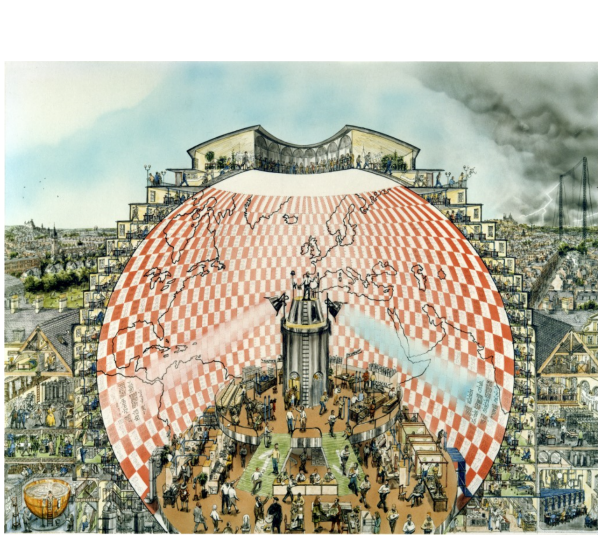
Lewis Fry Richardson (1881–1953)

Lewis Fry Richardson was a British mathematician and meteorologist who pioneered numerical weather prediction and introduced quantitative methods into the study of turbulence and conflict.

Contribution to numerical weather prediction:

Richardson was one of the first scientists to attempt weather forecasting by directly solving the equations governing atmospheric motion. In his 1922 book *Weather Prediction by Numerical Process*, he used observations from 20 May 1910 to calculate the future pressure field by hand using finite-difference methods. His forecast was wildly inaccurate, mainly because the calculation lacked adequate numerical filtering of the initial data. Nevertheless, it demonstrated that weather forecasting could be treated as a numerical problem, an idea confirmed by the first computer forecasts decades later.

He famously imagined a "forecast factory" in which thousands of human computers would work simultaneously on different parts of the atmosphere, anticipating modern numerical weather prediction. Richardson also introduced the Richardson number, which measures the competition between buoyancy and shear in stratified turbulence, and discovered the coastline paradox, in which measured coastline length increases as the measuring scale becomes smaller. During the First World War, Richardson served with the Friends' Ambulance Unit rather than in combat and later developed mathematical models of conflict and war. After the loss of the *Titanic*, he also proposed acoustic methods for detecting icebergs, anticipating principles later used in sonar.



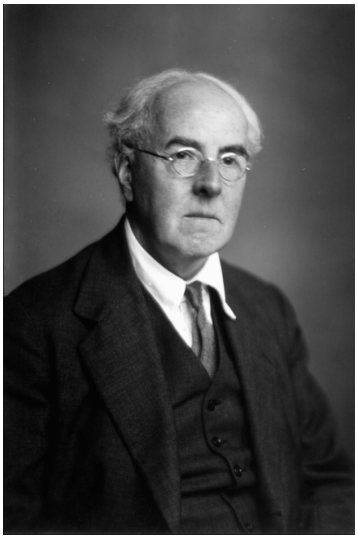


Figure 44: W*eather Forecasting Factory* by Stephen Conlin, based on the description by L.F. Richardson

Gilbert Walker (1868–1958)

Gilbert Walker was a British mathematician and meteorologist who discovered the Southern Oscillation and identified major statistical patterns in global atmospheric circulation.

Contribution to fluid mechanics:

Walker studied relationships between atmospheric pressure, winds, and rainfall using long-term weather observations. In the 1920s, he identified the Southern Oscillation, a large-scale variation in atmospheric pressure between the western and eastern tropical Pacific. This oscillation is closely linked to El Niño, and the combined ocean–atmosphere phenomenon is now known as ENSO(El Niño–Southern Oscillation). Walker also identified important pressure relationships associated with the North Atlantic Oscillation (NAO), describing how variations in the pressure difference between the Azores region and Iceland are related to changes in European weather and atmospheric circulation.

Walker's work demonstrated that weather patterns in widely separated regions can be statistically connected, providing an important foundation for the modern study of large-scale atmospheric circulation and climate variability.

Figure 45: Gilbert Walker

Theodore von Kármán (1881–1963)

Theodore von Kármán was a Hungarian-American mathematician and engineer who made major contributions to aerodynamics, turbulence, vortex shedding, and aerospace engineering.

Contribution to fluid mechanics: Von Kármán initially worked under Ludwig Prandtl at Göttingen, where he entered fluid mechanics and developed his interest in aerodynamics. After seeing an airplane demonstration in 1908, he worked on aerodynamic problems involving Zeppelin airships and later directed the Aeronautical Institute at Aachen, advancing wind-tunnel research and aircraft design. During the First World War, he served in the Austro-Hungarian military and worked on aeronautical research. His group developed tethered rotorcraft intended to replace observation balloons, making him an early pioneer of helicopter engineering. In fluid mechanics, he studied vortex shedding behind bluff bodies. Alternating vortices behind a cylinder form a repeating pattern called a von Kármán vortex street. His theoretical work helped establish the physics of wakes, flow separation, and flow-induced forces.

Von Kármán also developed the logarithmic law of the wall for turbulent boundary layers,

$$U(z) = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right),$$

where u_* is the friction velocity, z_0 is the roughness length, and $\kappa \approx 0.4$ is the von Kármán constant. The relation became fundamental for turbulent flows near surfaces and in the atmospheric boundary layer. In the 1930s, von Kármán moved to the United States and became a major figure in aeronautics and rocketry. At Caltech he helped develop rocket propulsion and established the laboratory that became the Jet Propulsion Laboratory.

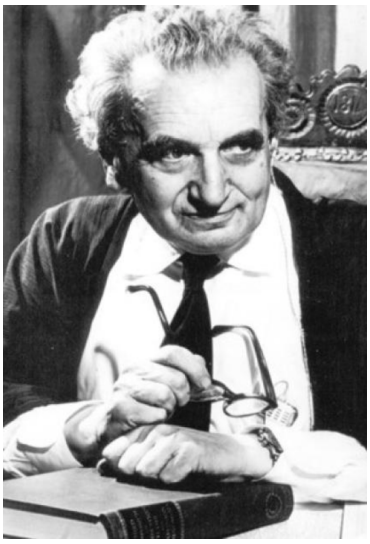


Figure 46: Theodore von Kármán

Geoffrey Ingram Taylor (1886–1975)

Geoffrey Ingram Taylor was a British mathematician and physicist who made fundamental contributions to turbulence, fluid transport, dimensional analysis, and the mechanics of explosions.

Contribution to fluid mechanics:

Taylor made major contributions to fluid mechanics, turbulence, and wave theory. He developed the statistical theory of turbulence by studying velocity fluctuations and correlations and introduced the Taylor microscale and Taylor's frozen-turbulence hypothesis, which relates measurements made over time to turbulent structures carried by the mean flow.

Taylor also studied many other fluid phenomena, including Taylor–Couette flow between rotating cylinders, Rayleigh–Taylor instability, Taylor dispersion in pipe flow, and the Taylor column in rotating fluids. His work connected fluid mechanics with meteorology and oceanography, including his early studies of atmospheric turbulence.

During both World Wars, Taylor applied fluid mechanics to military problems. In the Second World War he studied blast waves and underwater explosions and worked with John von Neumann on the theory of strong explosions. Their work contributed to the Taylor–von Neumann–Sedov description of blast waves. Taylor also used dimensional analysis and photographs from the Trinity nuclear test to estimate the explosive yield. His wide-ranging work helped establish modern statistical turbulence theory and demonstrated how fluid mechanics could be applied to atmospheric flows, ocean processes, aeronautics, and explosions.

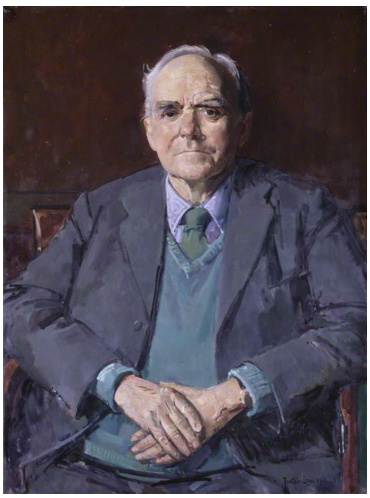


Figure 47: Geoffrey Ingram Taylor

Harald Ulrik Sverdrup (1888–1957)

Harald Ulrik Sverdrup was a Norwegian oceanographer and polar scientist who developed the theory linking wind stress to large-scale ocean circulation.

Contribution to physical oceanography:

Sverdrup was scientific director of Roald Amundsen's *Maud* expedition from 1918 to 1925, where he studied the Arctic Ocean, measuring depth, tides, currents, temperature, and salinity. His observations also contributed to the understanding of long waves over the Arctic shelf.

Later, at Scripps Institution of Oceanography, Sverdrup developed a theory linking wind stress to large-scale ocean circulation. A key ingredient was the *β-plane approximation*, which represents the variation of the Coriolis parameter with latitude:

$$f = f_0 + \beta y, \qquad \beta = \frac{df}{dy} = \frac{2\Omega \cos\phi}{R}.$$

Here, y is northward distance and β describes how the Coriolis effect changes across an ocean basin. Using this approximation, Sverdrup obtained the Sverdrup balance,

$$\beta M \rho_0 = \nabla_z \times \boldsymbol{\tau},$$

where M is the vertically integrated meridional transport and $\boldsymbol{\tau}$ is wind stress. This showed how wind-stress curl and the variation of the Coriolis parameter control large-scale ocean transport and became fundamental to the study of wind-driven ocean gyres.

Sverdrup also developed the critical-depth hypothesis for seasonal phytoplankton blooms. His 1942 textbook *The Oceans*, written with Martin Johnson and Richard Fleming, became a major foundation of modern oceanography.

Figure 48: Harald Ulrik Sverdrup

Carl-Gustaf Arvid Rossby (1898–1957)

Carl-Gustaf Arvid Rossby was a Swedish-American meteorologist and oceanographer who established the theory of large-scale atmospheric circulation and planetary Rossby waves.

Contribution to meteorology and fluid mechanics:

Rossby studied meteorology and oceanography under Vilhelm Bjerknes and later worked with oceanographic expeditions in Sweden. In 1928, he established the first American meteorological institute at MIT, where he initially studied atmospheric thermodynamics and air–sea interactions. During and after the Second World War, he played an important role in training military meteorologists and promoting mathematical and numerical approaches to weather forecasting.

At the University of Chicago, Rossby focused on large-scale atmospheric circulation. He explained the dynamics of jet streams and developed the theory of planetary-scale Rossby waves. His work showed that the variation of the Coriolis parameter with latitude is fundamental to large-scale motion, $f = f_0 + \beta y$. For a simple barotropic Rossby wave,

$$c = U - \frac{\beta}{k^2 + l^2},$$

so in a stationary background flow the waves propagate westward. Rossby's work provided a fundamental framework for understanding planetary-scale atmospheric circulation and ocean dynamics.

He also helped establish modern numerical weather prediction by applying mathematical models to electronic computers. His contributions connected atmospheric dynamics, oceanography, numerical forecasting, and the study of large-scale climate variability.

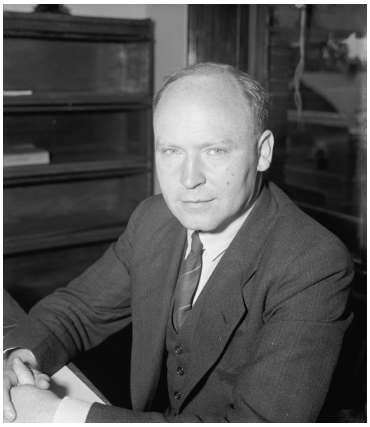


Figure 49: Carl-Gustaf Arvid Rossby

The Bergen School of Meteorology

The Bergen School of Meteorology, founded by Vilhelm Bjerknes, was one of the first organized centres for scientific weather forecasting. Its work transformed weather maps into tools for understanding and predicting the development of cyclones.

Jacob Bjerknes (1897–1975)

Jacob "Jack" Bjerknes developed the Norwegian cyclone model, explaining midlatitude cyclones as organized systems developing along a polar front between contrasting air masses. He showed how warm and cold fronts form and evolve around a cyclone, establishing a foundation of modern synoptic meteorology and weather forecasting. Jack later moved to the United States, where he became a leading figure in meteorology, studying the El Niño and helped planning of the Hiroshima and Nagasaki atomic bombings during WWII.

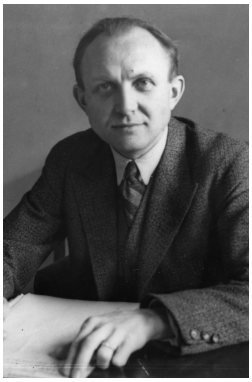
Halvor Solberg (1895–1974)

Halvor Solberg was a norwegian meteorologist who worked with Jacob Bjerknes on polar-front theory and helped explain how extratropical cyclones originate along the polar front, where warm and cold air masses meet. His work provided an important physical basis for the Norwegian cyclone model.

Tor Bergeron (1891–1977)

Tor Bergeron was a Swedish meteorologist who studied the later stages of cyclone development and explained how cyclones become occluded. When the faster cold front catches the warm front, the warm air is lifted from the surface and an occluded front forms, eventually weakening and dissipating the cyclone.

(a) Jacob Bjerknes



(b) Halvor Solberg

(c) Tor Bergeron

Figure 50: The Bergen School of Meteorology

Andrey Nikolaevich Kolmogorov (1903–1987)

Andrey Kolmogorov was a Soviet mathematician who developed the statistical theory of turbulence and the famous $-5/3$ energy spectrum.

Contribution to fluid mechanics:

Kolmogorov developed one of the most influential mathematical theories of turbulence. He proposed that at sufficiently small scales, turbulent motions become statistically universal and are governed primarily by the rate of kinetic-energy dissipation, ε, and the scale of the turbulent motion. His famous inertial-range energy spectrum is

$$E(k) \propto \varepsilon^{2/3} k^{-5/3},$$

where k is the wavenumber. The $(-5/3)$ power law is one of the most important signatures of the turbulent energy cascade.

In this picture, kinetic energy introduced into a flow at large scales is transferred through progressively smaller eddies. In the inertial range, this transfer is largely independent of the details of how the energy was originally introduced, until viscosity becomes important at the smallest scales and converts the kinetic energy into heat.

Kolmogorov's broader work in probability and stochastic processes also provided mathematical tools for describing random and fluctuating phenomena. His work on Markov processes, statistical prediction, and probability theory became important in the mathematical treatment of turbulent flows and other complex physical systems.

During World War II Kolmogorov contributed to the Soviet war effort by applying statistical theory to artillery fire, developing a scheme of stochastic distribution of barrage balloons intended to help protect Moscow from German bombers during the Battle of Moscow.

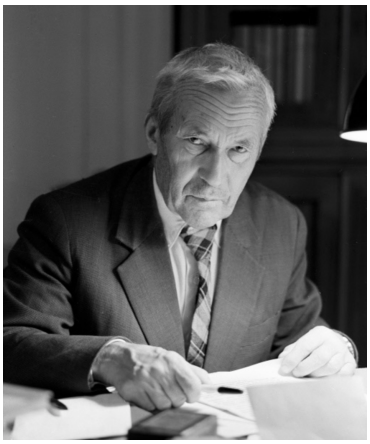


Figure 51: Andrey Nikolaevich Kolmogorov

Jule Gregory Charney (1917–1981)

Jule Gregory Charney was an American meteorologist who developed quasi-geostrophic theory and led the first successful numerical weather prediction experiment.

Contribution to meteorology:

Charney developed mathematical theories that simplified the atmospheric equations while retaining the dominant large-scale dynamics. He was a major contributor to quasi-geostrophic theory, which describes large-scale atmospheric flow in which the pressure-gradient force and Coriolis force are approximately balanced. The geostrophic balance can be written schematically as

$$f\mathbf{k} \times \mathbf{u}_g = -\frac{1}{\rho}\nabla_h p.$$

Charney also led the first successful numerical weather prediction experiment in 1950 together with Von Neumann. Using the newly developed electronic computer ENIAC, his group numerically integrated simplified atmospheric equations to forecast the evolution of a large-scale pressure pattern, altough the 24 hour forecast took 24 hours to compute.

This demonstrated that Richardson's vision of numerical weather prediction was computationally possible.

Figure 52: Jule Gregory Charney

Edward Norton Lorenz (1917–2008)

Edward Norton Lorenz was an American meteorologist who discovered deterministic chaos in atmospheric models and established the fundamental limits of long-range weather prediction.

Contribution to meteorology:

Lorenz studied the atmosphere through atmospheric circulation, numerical weather prediction, and nonlinear dynamical systems. In his work on general circulation, he developed the concept of available potential energy (APE), the part of atmospheric potential energy that can be converted into kinetic energy and drive large-scale circulation.

In the 1950s, Lorenz used computers to simulate simplified atmospheric models. In 1961, he discovered that a tiny change in a rounded initial value could produce a completely different weather evolution. His simplified convection model was

$$\begin{aligned} \frac{dx}{dt} &= \sigma(y - x), \\ \frac{dy}{dt} &= x(\rho - z) - y, \\ \frac{dz}{dt} &= xy - \beta z. \end{aligned}$$

Although deterministic, these equations exhibit sensitive dependence on initial conditions: small differences can grow rapidly. This revealed a fundamental limit to long-range weather prediction and became a foundation of chaos theory, later popularised as the "butterfly effect." Lorenz showed that limited predictability is inherent to the nonlinear dynamics of the atmosphere, helping establish the basis for modern ensemble forecasting.

Figure 53: Edward Norton Lorenz

Henry Melson Stommel (1920–1992)

Henry Melson Stommel was an American oceanographer who explained western intensification and the existence of strong western boundary currents.

Contribution to physical oceanography:

Stommel developed a theoretical explanation for the return flow required to close the large-scale ocean gyre predicted by Sverdrup's theory. He showed that when a simple linear drag term is included,

$$-\Gamma \mathbf{u},$$

the return flow becomes concentrated along the western boundary rather than being distributed symmetrically around the ocean basin. This produced the theoretical explanation for western intensification and for strong western boundary currents such as the Gulf Stream and Kuroshio.

Stommel's model showed that the north–south variation of the Coriolis effect causes the western boundary current to become much stronger and narrower than the corresponding eastern boundary flow. His work therefore solved a major problem left by Sverdrup's interior circulation theory and became a foundation of modern physical oceanography.

Together with Arnold Arons, Stommel later developed a theory of the deep ocean circulation involving sinking in high-latitude regions and deep western boundary currents. He also developed early models of thermohaline circulation, showing that the ocean circulation could possess more than one stable state.

Figure 54: Henry Melson Stommel

Walter Heinrich Munk (1917–2019)

Walter Heinrich Munk was an Austrian-American physical oceanographer who developed a comprehensive theory of wind-driven ocean circulation and made major contributions to waves, tides, and ocean acoustics.

Contribution to physical oceanography:

Munk extended Sverdrup's theory of wind-driven ocean circulation by solving the problem of how the interior transport returns to close the gyre. In his 1950 paper, he introduced lateral friction represented by a Laplacian diffusion of momentum,

$$A_H \nabla^2 \mathbf{u},$$

where A_H is a horizontal eddy-viscosity coefficient. This produced a narrow western boundary current that carries the return flow, but unlike the Stommel model, the current does not have its maximum velocity exactly at the coast. Instead, the velocity reaches a maximum slightly offshore. This became known as the Munk boundary-layer solution and provided a more realistic model of western boundary currents such as the Gulf Stream and Kuroshio. Munk also made major contributions to wave forecasting, tidal analysis, internal-wave theory, and ocean acoustic tomography. His wartime wave-forecasting work was particularly important because it allowed sea conditions to be predicted from wind speed, duration, and fetch, helping determine when amphibious operations, such as D-day, could be carried out safely.

Figure 55: Walter Heinrich Munk

Joseph Smagorinsky (1924–2005)

Joseph Smagorinsky was an American meteorologist who pioneered numerical climate modelling and developed the Smagorinsky subgrid-scale model for unresolved turbulence.
Contribution to atmospheric fluid dynamics:

Smagorinsky developed an important model for representing turbulent motions that are too small to be resolved by a numerical atmospheric model. A weather or climate model divides the atmosphere into grid cells. Large turbulent structures can be explicitly represented, but smaller turbulent eddies occur below the grid scale.

Smagorinsky represented their effect using an effective turbulent viscosity. In simplified form,

$$\nu_t \propto \Delta^2 |S|,$$

where Δ is the grid or filter scale and $|S|$ represents the magnitude of the resolved strain rate. This is the basis of the Smagorinsky subgrid-scale model.

The approach became extremely influential in numerical weather prediction, large-eddy simulation, and climate modelling because it provided a practical way of representing unresolved turbulence.

Figure 56: Joseph Smagorinsky